

TARIK BULUT

140 Smith Street, Darwin City, NT 0800

☎ 0424 512 965

✉ tarik.estaed@gmail.com

🌐 [linkedin.com/in/tarikbulut](https://www.linkedin.com/in/tarikbulut)

🐙 github.com/Estaed

Education

Charles Darwin University

Master of Information Technology (Artificial Intelligence) | GPA: 6/7

Nov 2024 – Nov 2026

Darwin, Australia

Firat University

Bachelor of Technology (Software Engineering) | GPA: 3.26/4

Aug 2019 – Jul 2023

Elazığ, Turkey

Experience

North Australia Centre for Autonomous Systems (NACAS)

Casual Professional

Mar 2026 – Present

Darwin, NT, Australia

- Developing **CV pipelines** and **AI models** for industrial applications, managing **data collection, preprocessing, and deployment**.
- Designing and optimizing **autonomous systems** for real-world environmental and **industrial monitoring**.
- Leveraging **Python** and **deep learning** to enhance **automated data analysis** accuracy and efficiency.

Procusale

Remote Intern / Volunteer

Nov 2021 – Nov 2023

Izmir, Turkey

- Developed cross-platform **web and mobile applications** using **Flutter, Firebase, and WordPress** for NGOs and law firms.
- Designed **UI/UX prototypes** in **Figma** and managed cloud infrastructure deployments on **DigitalOcean**.

Projects

FishNT: AI-Based Fish Classification (Master's Thesis) | Python, YOLO26, SAM 2.1, OpenCV

May 2026

- Developed a CV pipeline with **NT Government Fisheries** automating conveyor-belt fish detection, classification, and weight prediction.
- Engineered multi-stage **YOLO26-seg** and **YOLO26-cls** (7-class) architecture, achieving **99.6%** accuracy via 4-stream evidence fusion.
- Automated dataset amplification from 136 to 4,566 instances using **SAM 2.1** mask propagation.
- Executed calibrated pixel-to-real-world mapping, attaining **0.63 cm MAE** in length estimation for commercial species.

IT Code Fair Projects – Dark Sky NT & Kriol Translation | Python, PyTorch, Hugging Face

Nov 2025

- **Dark Sky NT:** Developed a tourism recommendation chatbot via multi-source pipelines, utilizing fine-tuned **multilingual BERT** (sentiment) and **LoRA-adapted Qwen** with **FAISS** (RAG).
- **Kriol Translation:** Built an end-to-end **NMT** system (**NLLB-200**) translating Kriol to English, utilizing low-resource data augmentation.
- **Achievements:** **Winner (AI) & Runner-Up (Data Science)** at CDU IT Code Fair 2025; Nominated for **NT Digital Excellence Awards 2025**.

Connect4 & League of Legends Wiki MCP | PyTorch, PPO, MCP, FastAPI, Selenium

Jul 2025 – Aug 2025

- Architected complex system integrations focusing on rapid prototyping and modular design using **AI-aided coding (Claude)**.
- Trained a **Connect4 RL agent** via **PPO** self-play in hybrid **CPU-GPU vectorized environments**.
- Built an **MCP server** querying real-time **LoL Wiki data** to **mitigate hallucinations** in AI tactical recommendations.

Doom - Mario AI | Python, Stable Baselines, Gym

Jan 2023

- Developed RL agents playing **Mario** and **Doom (1992)** using **PPO** and **DQN** algorithms.
- Authored a comparative performance analysis of RL algorithms in diverse environments (published on GitHub).

Technical Skills

Languages: Python, Java, C#, Dart, SQL, HTML/CSS

Libraries/Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, FastAPI, Hugging Face, Transformers

AI/ML Competencies: Computer Vision, RL (PPO/DQN), LLMs, RAG (FAISS), Fine-tuning (LoRA), NLP, Data Pipelines, APIs

Tools & Cloud: Git, Docker, VS Code, DigitalOcean, Firebase, Android Studio, Figma, Selenium, MCP Protocol, REST API

Honors, Languages & Interests

Honors: Nominated **NT Digital Excellence Awards 2025**; Winner (AI) & Runner-Up (Data Science) at **CDU IT Code Fair 2025**

Languages: Turkish (Native), English (Professional), German (Elementary)

Interests: Complex Strategy Games (**Europa Universalis IV-V**), League of Legends